Software Requirements Specification for Garbage Management System

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of the system that is being proposed is to develop garbage collection management system, version 1.0. The garbage collection management system is web based platform developed to streamline the efficient garbage collection process by enhancing communication and coordination among residents, garbage collectors, and municipal councils and promote eco friendly waste disposal practice

This document defines the functional and non functional requirements of the entire system, including user registration, real time vehicle tracking, garbage type detection through image upload, complaint management, scheduling, and administrative operations. The system aims to eliminate every single one of the problems that were caused by the manual garbage collection management and help the admins and residents to do a better garbage management.

1.2 Document Conventions

Category	**Description**
Font	Times New Roman
Size	12
Line & Paragraph Style	1.15
Style	Normal
Color	Black

1.3 Intended Audience and Reading Suggestions

The target audience of this document is who are involve in garbage management in government service sector. The municipal council officers, garbage collectors, garbage truck drivers, and resident around the country will find this document useful in understanding how they can be a part of this productive garbage collection management and to create a beautiful and clean nation. Each and every human living around municipal councils will be the main target audience who would want to dispose garbage in proper

way. This document will explain how they can use and navigate through the system to organize and manage their garbage efficiently.

1.4 Project Scope

The garbage management system is a web based platform developed to improve the efficiency, transparency, and sustainability of garbage collection services Currently garbage collection in many areas lacks proper scheduling and communication between municipal council and residents. Residents are often unaware of collection time, residents don't have knowledge about garbage types, and councils have limited tools to monitor collection progress and manage complaints.

This proposed system addresses these challenges by providing a centralized platform for all stakeholder

The detailed scope of the system is as follows:

Garbage management system is a critical and continuously evolving field that plays a major role in maintaining environment cleanliness and public health. It involves planning and coordinating the collection, transportation of garbage while minimizing environmental and health impacts.

At present, most garbage collection activities in Sri Lanka are managed manually. Municipal council often rely on outdated methods for recording and monitoring garbage truck movements, while residents remain unaware of the collection schedules. Complaints regarding missed pickups, overflowing bins, or irregular service must be filled in person, leading to delays and inefficiency. Data related to routes, drivers, and complaints are typically handled using paper based systems or basic tools such as spreadsheets, resulting in data loss, human error, and lack of transparency.

The aim of developing this proposed web based garbage collection management systems is to create an integrated digital platform that connects residents garbage collectors, and municipal councils. The system will automate the garbage collection process, enhance communication between stockholders, and promote sustainable waste management practices.

The detailed scope of the proposed system is as follows:

Admins/ Municipal council officers

- Register and create account for admin

- Track the garbage trucks
- Assign drivers, vehicles, & create bins
- Replay to complaints or take actions
- Conduct awareness campaign and post announcement or events for residents
- Send email notifications

Garbage truck drivers/ collectors

- Create profile and log into system
- Update bin collection routes and schedules
- Report operational issues
- Update complaints

Residents

- Create profile and log into system
- Upload image and separate it according to type
- Track the truck in real time and view collection schedule
- File a complaint & follow it
- View past complaint status
- Receive notifications

1.5 References

2. Overall Description

2.1 Product Perspective

The garbage collection management system is a new, self contained web based application design to replace the existing manual garbage collection management process used by many municipal councils. Currently, waste collection and monitoring rely heavily on physical records, phone calls, and direct communication between residents and municipal staff. This traditional approach leads to inefficient, delayed pickups, miscommunication, and lack of transparency.

The proposed garbage collection management system provides a centralized digital platform that integrates the key stakeholders, residents, garbage collectors, and municipal council administrators into one unified system. It is not a follow on or upgrade to an

existing software but an entirely new solution to automate, track and monitor the garbage collection process effectively.

The system operates as web application, accessible through standard web browsers on computers and mobile devices. It includes front end interface for users (residents, garbage collectors, and municipal council administrators) and a back end database for storing user data, collection reports, and complaint records.

Garbage collection management system interacts with external systems only for standard web functions such as location tracking. It is otherwise self contained and does not depend on external enterprise system

2.2 Product Features

The Garbage collection management system will provide a set of high level features that support residents, drivers, and municipal council administrators. These features are deigned to improve communication, transparency, and efficiency in waste collection operations

1.User registration and & profile management

- Residents, drivers, and admin officers can create and manage their profiles.
- Role based access ensures that users sees only the functionalities relevant to them.

2.Real time garbage truck tracking

- Residents can view the live location of the garbage truck.
- The municipal council can track garbage truck.
- Reduce uncertainty and prevent missed pickups.

3.Garbage collection scheduling

- Display collection schedules for residents.
- Admin can create, update, and manage garbage pickup schedules.

4.Image based garbage type detection

- Resident can upload an image of their garbage.
- The system provides information about the type of garbage.

5.Complaint management system

- Residents can report issues such as missed pickups, overflowing bins, or misconduct.

- Admins can view, manage, and take action on submitted complaints.
- Status updates keep residents informed. (Eg: pending, in progress, resolved)

6.Bin management

- Admin can create new bins, update bin status, assign bins to routes, and manage bin locations.
- Drivers can update bin status after collection. (Eg: emptied, overflowing, not accessible)

7.Route & driver assignment

- Admin can assign routes to trucks and allocate drivers.
- Management of daily operations such as task allocation.

8.Awareness & information sharing

- System provides recycling tips, health guidelines, and updates on environment events.
- Helps educate residents and promote sustainable waste management practices.
- System sends notifications regarding notices

2.3 User Classes and Characteristics

The system will be used by three primary user classes: Residents, Drivers/ Collectors, and Municipal administrators. Each class interacts with different system features based on their role, technical capability, and privilege level.

Residents

Residents are the general public and represent the largest user group of system. They primarily use the platform to track collection trucks, view their area's collection schedule, report issues, and identify garbage types by uploading images. Their technical expert is basic, so the interface for this user class should be simple and user friendly. Residents operate with the lowest privilege level but are considered the most important user class, as the system is designed to improve their experience, reduce complaints, and create a cleaner environment.

Drivers/ Garbage collectors

Drivers and garbage collectors use the system to perform their daily operational tasks. They log into view routes, update bin status, and confirm completion of pickups. Their technical knowledge is low to moderate, and the system should provide clear, minimal steps to

update daily activities. Driver use the system should provide clear, minimal steps to update daily activities. Drivers use the system frequently, mostly during working hours and require reliable, easy to navigate interfaces. Their privilege level is moderate, as they work with route data and bin updates but do not manage system wide operations. They are essential for keeping collection information accurate and up to date.

Municipal council administrators

Administrators are responsible for managing and controlling the entire system. They handle account creation for residents and drivers, manage collection schedules, assign routes, update bin details, monitor real time truck movements, process complaints and send email notifications to users. Admin users have high technical familiarity and frequently interact with the system throughout the day. They hold the highest privilege level and have full access to all operations and data. Administrators are a critical user class because the accuracy, reliability, and coordination of the whole system depend on their action and oversight.

2.4 Operating Environment

Hardware platform

The system is web based and cross platform, allowing it to run on desktop computers, laptops, tablets, and mobile devices used by residents, drivers and administrators. The server hosting the system should meet the minimum hardware requirements for running the backend, while client devices should support standard web browsing. Any hardware capable of running a modern browser with internet connectivity will be compatible.

Operating system

The system is designed to function on a wide range of operating systems, including windows, macOS, Linux, Android, and IOS. Since different user classes (Residents, Drivers, Administrators) access the system using different devices, the required OS version may vary, but all modern and commonly used versions are supported. The server environment may run on Linux or Windows server, depending on deployment choice.

Software components

The system relies on several software modules to ensure proper functionality. A relational database such as MySQL will be used for storing and managing user data, schedules

complaints, and tracking information. Backend development requires React, Node.js, Additional components include Google Maps API for real time truck tracking, TensorFlow.js for garbage type detection.

Peaceful Coexistence

All security tools such as antivirus software, firewalls, and network filters must operate without interfering with the system's functionality. The hosting environment and user devices should ensure compatibility with the application, allowing secure connections, map integration, image uploads, and database operations without disruption. Proper configuration is required so that these tools coexist smoothly with the garbage collection management system.

2. 5 Design and Implementation Constraints

The system must be developed using React.js for the frontend, Node.js for the backend, and MySQL as the database, limiting the use of alternative technologies. All communication must occur over secure HTTPS, and external services like Google Maps API and TensorFlow.js create dependencies on internet connectivity. The system requires a stable internet connection for real-time features such as truck tracking and image based garbage detection. It must also run smoothly on common user devices including smartphones and low spec computers, which limits the complexity of UI elements and processing tasks.

2.6 User Documentation

The following user documents will be provided with the garbage collection management system to support all user groups.

User manual

A complete digital user manual explaining system features, login procedures, navigation, and step by step instructions for performing key tasks such as submitting complaints, tracking trucks, and updating bin status.

Quick start guide

A short introductory guide for new users, including screenshots and simple steps to help them begin he system quickly.

2.7 Assumptions and Dependencies

It is assumed that all users including residents, drivers, and administrators will have stable internet access compatible devices to interact with the web based system. The system also relies on third party components such as map APIs for real time truck tracking, image processing tools for garbage type detection, meaning any changes or limitations in these service may affect system functionality. It is assumed that the municipal council will provide accurate route, bin, driver information, and will maintain the server environment required for hosting the application. The effectiveness of the system depends on drivers regularly updating bin status and following assigned routes. Additionally, the project depends on the availability of required software tools, such as the selected web framework and the database management system. Any changes in external services, infrastructure, or data availability may impact the final system performance or require adjustments to stated requirements.

3. System Features

3.1 User Registration & Profile Management

3.1.1 Description and Priority

Residents, Drivers, and Admin users can register for the system and create profiles. Users can manage and update their personal information through their profile dashboard.

Priority: High

3.1.2 Stimulus/ Response Sequences

Stimulus: The user clicks on the “Register” button.

Responses: The registration form is displayed.

Stimulus: The user submits the form with valid registration details.

Responses: System creates the user account, shows a success message, and redirect to the login page.

Stimulus: The user attempts to register with an already used email.

Responses: System displays an error message. “Email already exists”

3.1.3. Functional Requirements

1.The system must allow residents, drivers, and admins to fill the registration form.

2.The system must validate name, email, address, password and role type

3.The system must prevent registration with duplicate email addresses.

4.After successful registration, users must be redirected to the login page.

5.System must store registered user details securely.

3.2. Login to the System

3.2.1 Description and Priority

All system users (Residents, Drivers, Admin) can log into access their specific dashboards and system features.

Priority: High

3.2.2 Stimulus/ Response Sequences

Stimulus: The user clicks the “Login” button.

Response: Login page displayed.

Stimulus: User **e**nters valid login credentials and clicks submit.

Response: A success message is displayed and the user is redirected to the respective dashboard.

Stimulus: User inserts invalid login details for three consecutive attempts.

Response: Login is temporarily blocked and notification email is sent.

3.2.3 Functional Requirements

1.Residents, drivers, and admins must be able to fill the login form.

2.The system must validate username and password.

3.The system must redirect each user type to the correct dashboard after logging in

4.After 3 failed login attempts, the system must block login temporarily and send email notification.

3.3 Garbage Type Detection

3.3.1 Description and Priority

Residents can upload an image of garbage to identify the type and receive guidance on proper disposal

Priority: High

3.3.2 Stimulus/ Response Sequences

Stimulus: The resident clicks “Detect Garbage Type”

Response: Image upload interface displayed.

Simulus: Resident uploads a clear image and submits.

Response: System analyze and displays identified garbage type with disposal instructions.

Simulus: Residents uploads a blurry or unsupported image.

Response: System shows an error message: “Invalid image. Please upload clear image”

3.3.3 Functional Requirements

1. The system must allow users to upload garbage images.
2. The system must analyze the image and identify it.
3. The system must display correct disposal instructions.
4. The system must store detection logs.
5. The system must show an error if the image is unclear or invalid.

3.4 Garbage Truck Tracking

3.4.1 Description and priority

Residents and Admins can track garbage trucks in real time and view estimated arrival times

Priority: High

3.4.2 Stimulus/ Response Sequences

Stimulus: The user selects “Track Garbage Truck”

Response: System loads map and fetches truck GPS data.

Stimulus: Truck location updates in real time.

Responses: System displays live truck movement and

Stimulus: GPS device is unreachable or offline.

Responses: System displays “tracking unavailable” message.

3.4.3 Functional Requirements

1. System must collect GPS coordination of garbage trucks.
2. System must update the map in real time.
3. System must show estimated arrival time.
4. System must notify if GPS tracking fails.
5. System must limit access to authorized users only.

3.5 Garbage Collection Scheduling

3.5.1 Description and Priority

System provides daily and weekly garbage collection schedules for residents and allows admin to modify schedules.
Priority: High

3.5.2 Stimulus/ Response Sequence

Stimulus: Residents clicks “Schedule”

Response: System displays daily schedules.

Stimulus: Admin edits the schedule.

Response: Schedule updates and publishes instantly.

3.5.3 Functional Requirements

1. System must show garbage collection schedules based on resident location.
2. System must allow admin to edit schedules.
3. System must update schedules in real time.
4. System must notify residents about schedule changes
5. .System must store schedule history.

3.6 Complaint Management

3.6.1 Description and Priority

Allow residents to submit complaints regarding missed pickups, overflowing bins, or driver behavior, and allows admin to manage resolutions.

Priority: High

3.6.2 Stimulus/ Response Sequences

Stimulus: Resident click “Submit Complaint”.

Response: Complaint form displayed.

Stimulus: Resident submit valid complaint.

Response: System records complaints.

Stimulus: Admin update complaint status.

Response: Resident receives updates on complaints.

3.6.3 Function Requirements

1. System must allow residents to submit complaints.
2. System must validate complaint information.
3. System must allow admin to change complaint status.
4. System must notify residents when complaint progress change.
5. System must store all complaint records.

3.7 Driver Route & Bin Status Management

3.7.1 Description and Priority

Drivers can view and assigned routes and update the status of garbage bins during collection.

Priority: High

3.7.2 Stimulus/ Response Sequences

Stimulus: Driver logs in and open “Assigned Routes”.

Response: System displays bins and locations.

Stimulus: Driver mark bin as “Collected”.

Response: System records the status.

3.7.3 Function Requirements

1. System must display assigned routes to drivers
2. System must allow drivers to update bin status.
3. System must notify admin for issues like overflow.

3.8 Admin Management (Bins Drivers, Routes, Reports)

3.8.1 Description and Priority

Admin manages bins, drivers, routes trucks, complaints, and system operations.

Priority: High

3.8.2 Stimulus/ Response Sequences

Stimulus: Admin select “Manage Bins”

Response: System displays all bin locations.

Stimulus: Admin assign drivers, routes and bins

Response: System display update new assignments**.**

3.8.3 Function Requirements

1. System must allow admin to add, edit, and delete bins.
2. System must allow admin to assign drivers and trucks.
3. System must allow admins to manage drivers and trucks.

3.9 Public Awareness & Notifications

3.9.1 Description and Priority

System displays recycling guidelines, health information, and sends notifications related to garbage management

Priority: Medium

3.9.2 Stimulus/ Response Sequences

Stimulus: Resident opens “Awareness page”

Responses: System displays articles, tips, and guidelines**.**

Stimulus: Admin posts an event/ announcement.

Responses: System updates awareness page and send notifications.

3.9.3 Functional Requirements

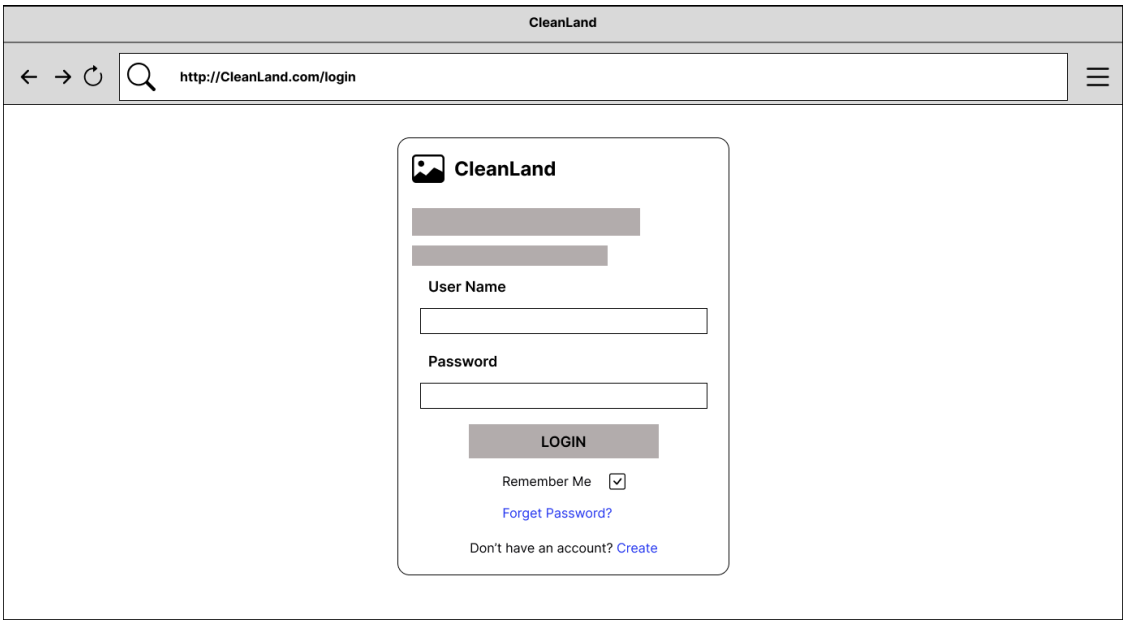
1. System must display segregation and recycling guidelines**.**
2. System must allow admin to upload awareness content.
3. System send notifications for important updates.
4. System must store all published content

4. External Interface Requirements

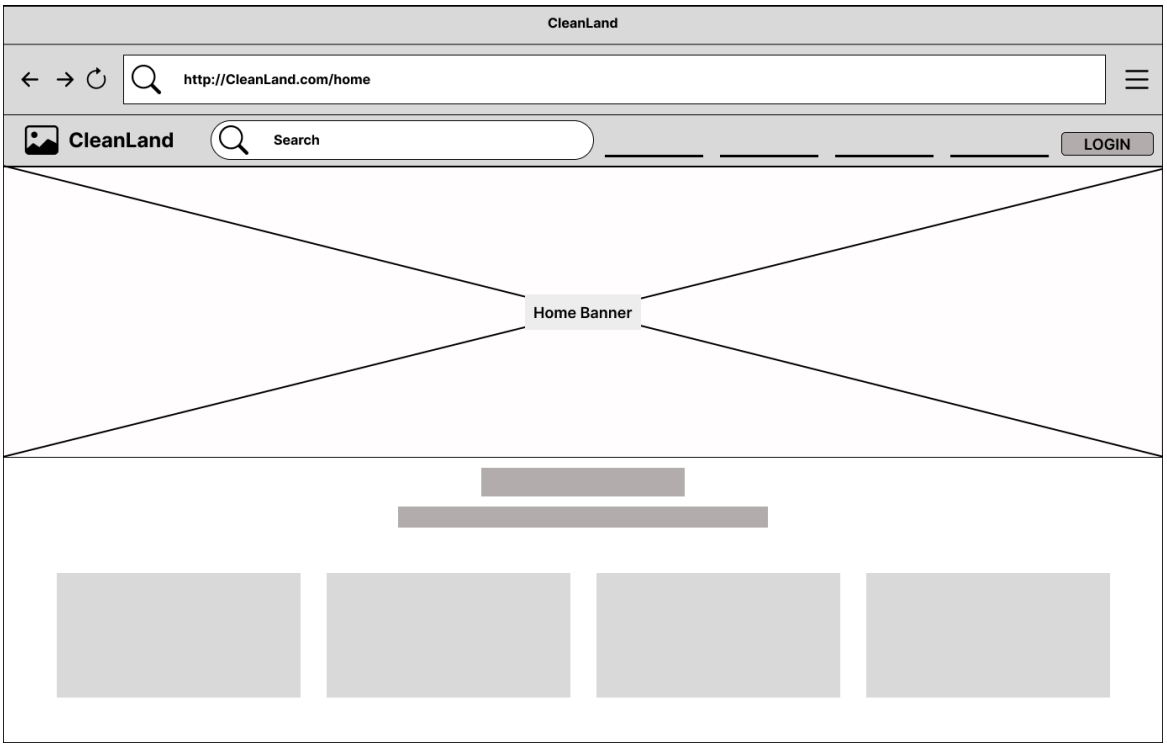
4.1 User Interfaces

The following are some wireframes that shows how the interfaces of the application shall appear:

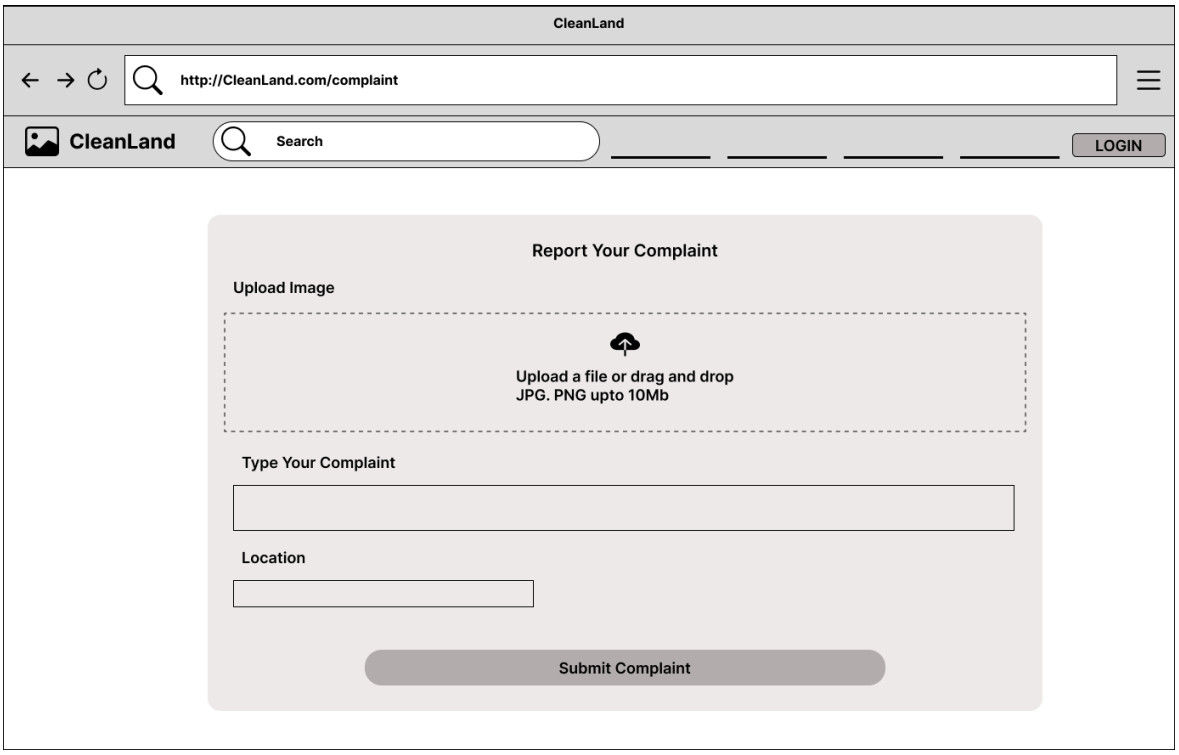
Login - This screen is used to login into the system



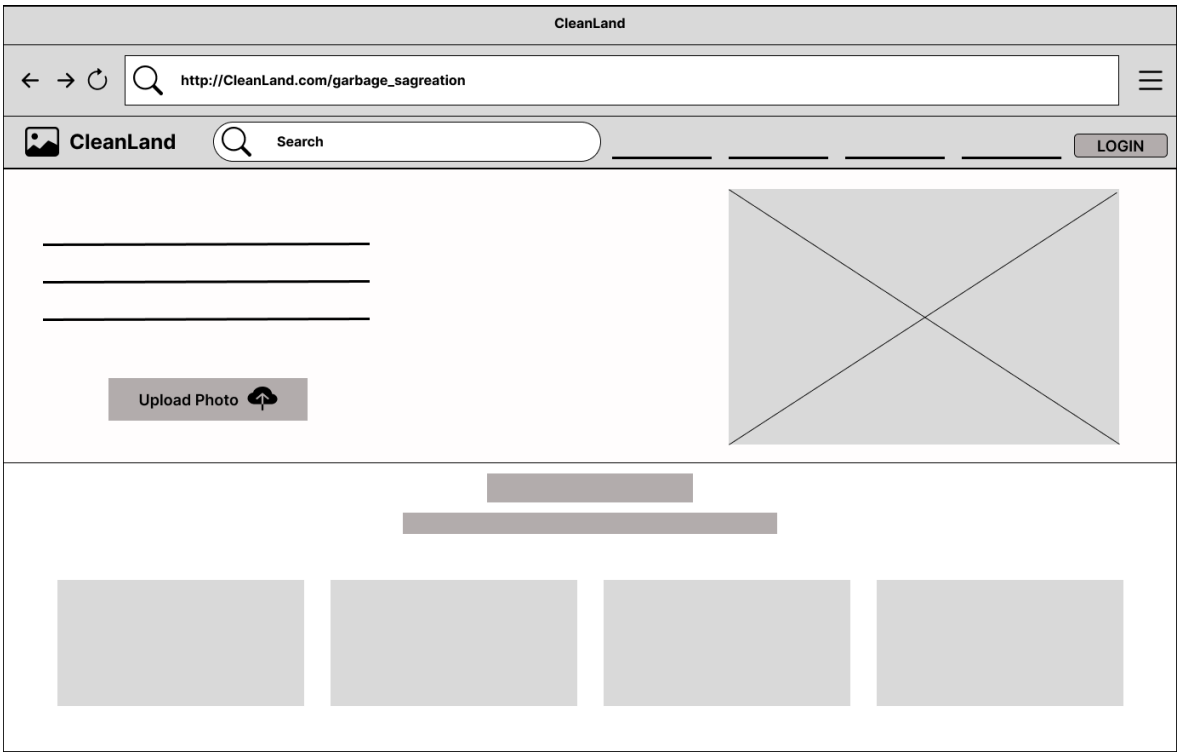
Home Page


CleanLand
http://CleanLand.com/home
CleanLand
Search
LOGIN
Home Banner

Complaint Page



Type Detection Page



4.2 Hardware Interfaces

This system is a web based application, so smartphone, laptop, or desktop computer are the hardware devices that are needed to connect to the system additionally, the system will be able to work with a wide variety of operating system as this is developed to be cross platform. It will be hosted in a server after deployment

4.3 Software Interfaces

Software and script Components	**Name and Version**	**Description**
Operating system	The system is compatible with Windows, macOS, Linux, Android, and iOS through a web browser.	The system will be developed to run on all the operating system
Web browser	Google chrome, Mozilla Firefox	Supported browsers used by residents, drivers and admins to access the system.
Backend	Node.js- Node v24.x	Handles server side logic including authentication, routing, schedule management, complaint handling, and image processing request.
Database	MySQL- v8.4	Stores user accounts schedules, complaints, bin data, and configuration details**.**
Frontend	React.js- v19.2.0	Used to build the interactive user interface, dashboards, forms, tracking views, and all client-side components.
Mapping Service	Google maps	Provides map rendering.
Image Processing	TensorFlow.js	Performs garbage type detection from uploaded images.

4.4 Communications Interfaces

Communication Function	Description
Email	The system sends email notifications to users.
Browser	Users access the system through a web browser. The frontend is built using React.js and works on Chrome, Firefox, Safari, and Edge.
Electronic Forms	The system provides electronic forms for registration, login, complaints, and other actions. Forms send data to the backend through secure REST API calls.
Mapping communication	The system connects to Google Maps to load maps and show truck locations via secure HTTPS calls.
Database communication	Node.js communicates with the MySQL database to store and retrieve all system data using secure queries.
GPS tracking data	Garbage truck location data is sent securely to the backend and displayed on the frontend in real time.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The system should operate smoothly under heavy usage and support multiple users at the same time without slowing down. All main functions, including loading pages, viewing schedules, and tracking garbage trucks should respond within 5 seconds. The server setup must be strong enough to handle a high number of requests from residents, drivers and administrators during peak hours.

5.2 Safety Requirements

The system must protect all stored information and ensure that data is not lost or damaged. Regular backups should be maintained so that older records can be restored if needed. Sensitive operational pages or functions should be accessible only through secure, restricted links to keep the system safe from unauthorized access.

5.3 Security Requirements

The system should ensure that user data is kept secure at all times. This includes using protected storage methods and controlling who can access different parts of the system. Only authorized users should be able to log in and perform specific actions, ensuring that personal and operational information remains safe from misuses.

5.4 Software Quality Attributes

As the target system is a web-based system and supports any operating system. It is maintainable, adaptable, available, flexible, reliable, and usable.